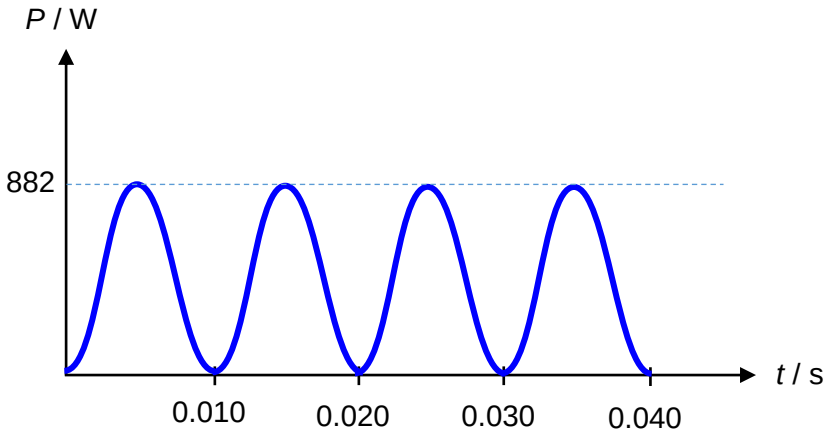


8	(a)		The steady direct voltage value which provides the same power / energy dissipation as the alternating voltage.	1
	(b)		$\langle V^2 \rangle = [(4^2 \times 2) + (2^2 \times 4)] / 10$ $= 4.8$ $V_{\text{rms}} = \sqrt{4.8}$ $= 2.19 = 2.2 \text{ V}$ Mean power = $V_{\text{rms}}^2 / R = 2.19^2 / 25 = 0.192 \text{ W}$	1 1 1
	(c)		For a given power, higher voltage means lower current in cable. Lower current in cable will result in lower power lost in cable.	1 1
	(d)	(i)	$\frac{V_s}{V_p} = \frac{N_s}{N_p}$ $V_s = 70 \times 15 = 1050 \text{ V}$	1

		(ii)	 Labelling of correct P_0 value – 1 mark $P_0 = V_0^2 / R = (\sqrt{2} \times 1050)^2 / 2500 = 882 \text{ W}$ Correct shape – 1 mark	2
8	(a)	(i)	Wavelength $\lambda = 2.4 \text{ cm}$	